

# Tic-Tac-Toe Game with AI Player

## Abstract

This project develops a Tic-Tac-Toe game where a human player competes against an AI opponent. The AI uses the Minimax Algorithm to select the optimal move, making it extremely difficult to defeat.

## Introduction

Artificial Intelligence enables computers to make decisions similar to humans. This project demonstrates AI decision-making through a classic Tic-Tac-Toe game.

## Problem Statement

Develop an intelligent Tic-Tac-Toe game that allows a human player to compete against an AI capable of making optimal decisions.

## Objectives

- Build a Human vs AI game.
- Implement Minimax Algorithm.
- Detect win, loss and draw conditions.
- Understand game theory concepts.

## Software Requirements

Python, Tkinter (GUI), VS Code, Windows/Linux OS.

## Hardware Requirements

Intel i3 or above, 4GB RAM, 1GB free storage.

## System Architecture

Human Player → Game Board → Minimax Algorithm → Best Move Selection → AI Move → Result Display

## Modules

1. User Interface Module
2. Game Logic Module
3. AI Engine Module
4. Result Evaluation Module

## Working Procedure

The player enters a move. The game validates it. The AI evaluates all possible future states using Minimax and chooses the best move. The game continues until a win or draw occurs.

## Minimax Algorithm

Minimax recursively evaluates all possible moves. AI tries to maximize score while the opponent tries to minimize score. Scores: AI Win = +10, Human Win = -10, Draw = 0.

## Algorithm Steps

1. Check terminal state.
2. Generate possible moves.
3. Recursively evaluate moves.
4. Select move with highest score.
5. Update board.

## Sample Board

```
X | O | X
O | X | O
O | | X
```

## Advantages

- Intelligent gameplay
- Demonstrates AI concepts
- Improves problem-solving skills
- Easy to understand and implement

## Applications

Game development, AI education, algorithm learning, academic projects.

## Future Scope

Online multiplayer support, difficulty levels, reinforcement learning, voice-controlled gameplay.

## Testing

Unit testing for move validation, win detection, draw detection and AI move generation.

## Result

The AI successfully predicts optimal moves and provides a challenging gaming experience.

## Conclusion

The project demonstrates practical implementation of AI using the Minimax Algorithm and helps students understand game theory and decision-making systems.

## References

1. Artificial Intelligence by Stuart Russell.
2. Python Documentation.

### 3. Minimax Algorithm Tutorials.